

Chapter 5, Section 4

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1. The linear system of conics in $\mathbb{P}^2$ with two assigned base points P_1 and P_2 (4.1) determines a morphism ψ of X' (which is $\mathbb{P}^2$ with P_1 and P_2 blown up) to a nonsingular quadric surface Y in $\mathbb{P}^3$, and furthermore X' via ψ is isomorphic to Y with one point blown up.

Proof. We may take the two points to be $P_1 = (1 : 0 : 0)$ and $P_2 = (0 : 1 : 0)$. Then the vector space $V \subseteq H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2))$ corresponding to $\mathfrak{d}$ is spanned by

$$y_0 = x_1x_2, \quad y_1 = x_0x_2, \quad y_2 = x_0x_1, \quad y_3 = x_2^2,$$

where y_i are the homogenous coordinates of $\mathbb{P}^3$. Therefore, the points in the image of ψ satisfies the equation

$$y_0y_1 - y_2y_3 = 0,$$

which corresponds to a nonsingular quadric surface $Y \subset \mathbb{P}^3$.

Let L_i be the strict transform of P_i via π , let L be the line through P_1 and P_2 , which is defined by $x_2 = 0$ in $\mathbb{P}^2$, and let $\tilde{L}$ be the strict transform of L in X' .

$$\begin{array}{ccccc} & X' & & & \\ \pi \swarrow & & \searrow \psi & & \\ \mathbb{P}^2 & \xrightarrow{\varphi} & Y & \hookrightarrow & \mathbb{P}^3 \end{array}$$

□

2. Let φ be the quadratic transformation of (4.2.3), centered at P_1, P_2, P_3 . If C is an irreducible curve of degree d in $\mathbb{P}^2$, with points of multiplicity r_1, r_2, r_3 at P_1, P_2, P_3 , then the strict transform C' of C by φ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at Q_1 , $d - r_1 - r_3$ at Q_2 , and $d - r_1 - r_2$ at Q_3 . The curve may have arbitrary singularities.

Proof. We have a commutative diagram

$$\begin{array}{ccc} & X & \\ \pi \swarrow & & \searrow \psi \\ \mathbb{P}^2 & \xrightarrow{\varphi} & \mathbb{P}^2 \end{array}.$$

Let H be a hyperplane section of $\mathbb{P}^2$, so that φ is given by the linear system $\mathfrak{d} = |2H - P_1 - P_2 - P_3|$, and let E_1, E_2, E_3 be the exceptional divisors of X via the blowing-up of $\mathbb{P}^2$ at P_1, P_2, P_3 , respectively. By (3.6), the strict transform $\tilde{C}$ of C by π is linearly equivalent to

$$\tilde{C} = \pi^*C - r_1E_1 - r_2E_2 - r_3E_3.$$

Since ψ is given by the linear system $\mathfrak{d}' = |D|$ with $D = 2\pi^*H - E_1 - E_2 - E_3$, φ is a birational map such that $C' = \varphi(C) = \psi(\tilde{C})$, and $C \sim dH$ in $\mathbb{P}^2$, the degree of C' in $\mathbb{P}^2$ can be computed using (Ex. 1.2) and (3.2)

$$\begin{aligned} \tilde{C}.D &= (d\pi^*H - r_1E_1 - r_2E_2 - r_3E_3).(2\pi^*H - E_1 - E_2 - E_3) \\ &= 2d(H^2) + r_1(E_1)^2 + r_2(E_2)^2 + r_3(E_3)^2 \\ &= 2d - r_1 - r_2 - r_3. \end{aligned}$$

For each $i \neq j$, let L_{ij} be the line which goes through P_i and P_j . By (3.6), for each $k = 1, 2, 3$, the multiplicity of C' at Q_k is the intersection of $\tilde{C}$ with the strict transform $\tilde{L}_{ij}$ of L_{ij} with $i, j \neq k$ via π (4.2.3), which by (3.6) is linearly equivalent to

$$\tilde{L}_{ij} = \pi^* L_{ij} - E_i - E_j.$$

Again, we have $L_{ij} \sim H$ in $\mathbb{P}^2$, so

$$\begin{aligned} \tilde{C} \cdot \tilde{L}_{ij} &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3) \cdot (\pi^* H - E_i - E_j) \\ &= d + r_i(E_i)^2 + r_j(E_j)^2 \\ &= d - r_i - r_j. \end{aligned}$$

Hence, C' has points of multiplicity $d - r_i - r_j$ at Q_k . $\square$

3. Let C be an irreducible curve in $\mathbb{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform C' of C has only *ordinary* singularities, i.e., multiple points with all distinct tangent directions.

Proof. Consider the cuspidal cubic curve in $\mathbb{P}^2$ defined by $y^2 = x^3$. $\square$

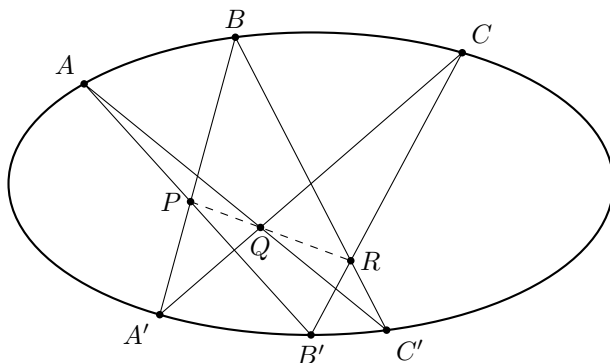
4. (a) Use (4.5) to prove the following lemma on cubics: If C is an irreducible plane cubic curve, if L is a line meeting C in points P, Q, R , and L' is a line meeting C in points P', Q', R' , let P'' be the third intersection of the line PP' with C , and define Q'', R'' similarly. Then P'', Q'', R'' are collinear.
(b) Let P_0 be an inflection point of C , and define the group operation on the set of regular points of C by the geometric recipe “let the line PQ meet C at R , and let $P_0 R$ meet C at T , then $P + Q = T$ ” as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

Proof.

- (a)
(b)

$\square$

5. Prove Pascal's theorem: if A, B, C, A', B', C' are any six points on a conic, then the points $P = AB' \cdot A'B$, $Q = AC' \cdot A'C$, and $R = BC' \cdot B'C$ are collinear.



6. Generalize (4.5) as follows: given 13 points $P_1, \dots, P_{13}$ in the plane, there are three additional determined points P_{14}, P_{15}, P_{16} , such that all quartic curves through $P_1, \dots, P_{13}$ also pass through P_{14}, P_{15}, P_{16} . What hypotheses are necessary on $P_1, \dots, P_{13}$ for this to be true?
7. If D is any divisor of degree d on the cubic surface (4.7.3), show that

$$p_a(D) \leq \begin{cases} \frac{1}{6}(d-1)(d-2) & \text{if } d \equiv 1, 2 \pmod{3} \\ \frac{1}{6}(d-1)(d-2) + \frac{2}{3} & \text{if } d \equiv 0 \pmod{3}. \end{cases}$$

Show furthermore that for every $d > 0$, this maximum is achieved by some irreducible nonsingular curve.

8. Show that a divisor class D on the cubic surface contains an irreducible curve $\iff$ it contains an irreducible nonsingular curve $\iff$ it is either (a) one of the 27 lines, or (b) a conic (meaning a curve of degree 2) with $D^2 = 0$, or (c) $D.L \geq 0$ for every line L , and $D^2 > 0$.
9. If C is an irreducible non-singular curve of degree d on the cubic surface, and if the genus $g > 0$, then

$$g \geq \begin{cases} \frac{1}{2}(d-6) & \text{if } d \text{ is even, } d \geq 8, \\ \frac{1}{2}(d-5) & \text{if } d \text{ is odd, } d \geq 13, \end{cases}$$

and this minimum value of $g > 0$ is achieved for each d in the range given.

10. A curious consequence of the implications (iv) $\implies$ (iii) of (4.11) is the following numerical fact: Given integers $a, b_1, \dots, b_6$ such that $b_i > 0$ for each i , $a - b_i - b_j > 0$ for each i, j , and $2a - \sum_{i \neq j} b_i > 0$ for each j , we must necessarily have $a^2 - \sum b_i^2 > 0$. Prove this directly (for $a, b_1, \dots, b_6 \in \mathbb{R}$) using methods of freshman calculus.
11. *The Weyl Groups.* Given any diagram consisting of points and line segments joining some of them, we define an abstract group, given by the generators and relations, as follows: each point represents a generator x_i . The relations are $x_i^2 = 1$ for each i ; $(x_i x_j)^2 = 1$ if i and j are not joined by a line segment, and $(x_i x_j)^3 = 1$ if i and j are joined by a line segment.

- (a) The *Weyl group* $\mathbf{A}_n$ is defined using the diagram



of $n - 1$ points, each joined to the next. Show that it is isomorphic to the symmetric group Σ_n as follows: map the generators of $\mathbf{A}_n$ to the elements $(12), (23), \dots, (n-1, n)$ of Σ_n , to get a surjective homomorphism $\mathbf{A}_n \rightarrow \Sigma_n$. Then estimate the number of elements of $\mathbf{A}_n$ to show in fact it is an isomorphism.

- (b) The *Weyl group* $\mathbf{E}_6$ is defined using the diagram



Call the generators $x_1, \dots, x_5$ and y . Show that one obtains a surjective homomorphism $\mathbf{E}_6 \rightarrow G$, the group of automorphisms of the configuration of 27 lines (4.10.1), by sending $x_1, \dots, x_5$ to the permutations $(12), (23), \dots, (56)$ of the E_i , respectively, and y to the element associated with the quadratic transformation based at P_1, P_2, P_3 .

- (c) Estimate the number of elements in $\mathbf{E}_6$, and thus conclude that $\mathbf{E}_6 \cong G$.
12. Use (4.11) to show that if D is any ample divisor on the cubic surface X , then $H^1(X, \mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).
13. Let X be the Del Pezzo surface of degree 4 in $\mathbb{P}^4$ obtained by blowing up 5 points of $\mathbb{P}^2$ (4.7).
- (a) Show that X contains 16 lines.
- (b) Show that X is a complete intersection of two quadric hypersurfaces in $\mathbb{P}^4$ (the converse follows from (4.7.1)).
14. Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbb{P}^3$ with $d = 8, g = 6, 7; d = 9, g = 7, 8, 9; d = 10, g = 8, 9, 10, 11$. Combining with (IV, §6), this completes the determination of all possible g for curves of degree $d \geq 10$ in $\mathbb{P}^3$.
15. Let $P_1, \dots, P_4$ be a finite set of (ordinary) points of $\mathbb{P}^2$, no 3 collinear. We define an *admissible transformation* to be a quadratic transformation (4.2.3) centered at some three of the P_i (call them P_1, P_2, P_3). This gives a new $\mathbb{P}^2$, and a new set of r points, namely Q_1, Q_2, Q_3 , and the images of $P_4, \dots, P_r$. We say that $P_1, \dots, P_r$ are in *general position* if no three are collinear, and furthermore after any finite sequence of admissible transformations, the new set of r points also have no three collinear.

- (a) A set of 6 points is in general position if and only if no three are collinear and not all six lie on a conic.

- (b) If $P_1, \dots, P_r$ are in general position, then the r points obtained by any finite sequence of admissible transformations are also in general position.
 - (c) Assume the ground field k is uncountable. Then given $P_1, \dots, P_r$ in general position, there is a dense $V \subseteq \mathbb{P}^2$ such that for any $P_{r+1} \in V$, $P_1, \dots, P_{r+1}$ will be in general position.
 - (d) Now take $P_1, \dots, P_r \in \mathbb{P}^2$ in general position, and let X be the surface obtained by blowing up $P_1, \dots, P_r$. If $r = 7$, show that X has exactly 56 irreducible nonsingular curves C with $g = 0$, $C^2 = -1$, and that these are the only irreducible curves with negative self-intersection. Ditto for $r = 8$, the number being 240.
 - (e) For $r = 9$, show that the surface X defined in (d) has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$.
- 16.** For the *Fermat cubic surface*. $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface?